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1 March 1955

MILITARY STANDARD

WEIGHT AND BALANCE DATA REPORTING FORMS FOR GUIDED MISSILES AND SPACE LAUNCH VEHICLES

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DEPARTMENT OF DEFENSE

WASHINGTON, D.C. 20360

Weight and Balance Data Reporting Forms for Guided Missiles and Space Launch Vehicles

1. This standard has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force, effective 27 May 1963.

2. Recommended corrections, additions, or deletions should be addressed to Bureau of Naval Weapons, Navy Dept., Washington, D.C., 20360.

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FOREWORD

This revision to MIL-STD-176 dated 1 March 1955 is based on the cooperative efforts of the Military Services, Weight Engineers of the Missile Industry, representatives of the National Aeronautics and Space Administration, the National Committee of the Society of Aeronautical Weight Engineers and the Aero Space Industries Association.

In general, this standard provides for the mass properties reporting of the complete spectrum of small and large missiles and space launch vehicles. These reported data provide the technical background for weight control engineering, weight optimization studies, the derivation of proposal analysis techniques and provides inputs to industrial planning reports for measuring the national industrial effort.

The standard follows the general philosophy used in MIL-STD-254 Weight and Balance Data Reporting Forms for Aircraft and MIL-STD-451 Weight and Balance Report Forms for Rotocraft and is adaptable to either manual or machine method preparation.

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WEIGHT AND BALANCE DATA REPORTING FORMS FOR GUIDED MISSILES AND SPACE LAUNCH VEHICLES

1. SCOPE

1.1 Scope. This document establishes weight statements and forms for procurement of weight and balance data for guided missiles, PARTS I, II and III hereof, states the principles followed in the formulation of these statements and forms, furnishes instructions where necessary for uniform compilation of the required weight and descriptive data, and provides a glossary of terms for which definitions may not be self evident on the weight statements and forms. The forms may be used for large and small missiles as well as booster vehicles employed for launching satellite or reentry payloads.

1.2 Concepts. The detail and summary weight statements have been predicated on the following basic concepts:

(a) The primary purposes of the weight data are to provide information in the most advantageous form for development and improvement of guided missiles weight estimating and optimization methods and for mass characteristics control during design, construction and operational use of these vehicles. Such data and methods serve the significant functions of bringing avoidable adverse trends to light at an early state, pointing out areas in which design weight control can be most fruitful, establishing realistic weight goals, evaluating on a rational basis the weight costs of design features during evaluation of new configurations, determining the deviations (from planned weight or mass characteristics) which occur under actual test or operating conditions, and providing timely controlled weight input to flight trajectory planning for changes during design and operation.

(b) The secondary purpose of the weight data is to provide reasonably detailed information for engineering checking and reference purposes.

1.3 Principles. The following principles have been followed to implement the above basic concepts:

(a) The total missile weight is subdivided into functional and identifiable divisions that best serve the basic concepts for both large and small missiles. The missile is subdivided into identifiable steps and/or sections as shown in Figure 1 as an example. This breakdown is accomplished by the use of vertical columns in the reporting forms which must be properly identified at the top of each column. This makes it possible to handle multistage vehicles and also to give sectionalized breakdowns of smaller missiles. It is permissible to use the vertical columns for sectional breakdowns within a step of a multistage vehicle by using more than one column for a single step. However, once a particular arrangement of columns has been established dividing the vehicle into steps and/or sections, the same arrangement must be used throughout the form for that particular missils

(b) Within the step and/or section breakdown, the weight has been broken down into functional and identifiable major and minor components. Insofar as practicable, all tangible and readily identifiable components and provisions for a specific function or installation have been allocated to the functional group, regardless of geographic location and physical method of attachment. Where compromise of this principle was necessary because of multi-purpose function, items have been allocated geographically with as complete functional isolation and identification as possible. Examples of multipurpose items are doors and panels which serve as closures for a structural surface as well as a particular installation function, and items which, because of their integral nature, make quantitative allocation impractical.

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(c) The importance of consistent allocation of weights by all contractors in completing the detail statement has been recognized. The amplifying instructions herein are furnished to insure consistency in allocation to the greatest extent practicable.

2. REFERENCED DOCUMENTS

2.1 The following standard forms a part of this standard to the extent specified herein:

STANDARD

Military

MIL-STD-12B-Abbreviations for Use on Drawings.

3. DEFINITIONS AND ABBREVIATIONS

Term	Definition
Step	The mass of that portion of a vehicle which may be physically separated from other portions by disconnecting at staging interfaces. (See Fig. 1)
Stage	The mass of that portion of a vehicle consisting of a Step plus all Steps above. (See Fig. 1)
Useful Payload	The mass of scientific instruments, cargo, weapons, etc., delivered to their destinations.
Carried Mass	The mass of that portion of a vehicle above the interface of the subject Step. The Carried Mass of Step 1 is Stage 2. (See Fig. 1)
Weight Empty	Those items that make up a basic empty vehicle weight. Does not include any fluids or lubricants.
Mission Load Fixed	Those items that must be added to an empty vehicle weight for flight but are not normally expendable; and those items that are aboard for a specific mission but not necessarily aboard for all flights.

ABBREVIATIONS—When abbreviations are necessary for write-ins or in machine tabulation methods, the abbreviations of Military Standard MIL-STD-12B shall be used wherever space permits. The following list summarizes acceptable abbreviations commonly used for weight data or used in PARTS, I, II, and III of this standard. Where more than one abbreviation is shown, the first given is preferred.

accessory—access., acc

adjust—adj

aerodynamic—aerodyn, aero
alighting—alight
ammunition—amm, ammo
antennas—ant
arresting—arg
attachment—att
auxiliary—aux

booster—bstr
bracket-brkt
brake-bk

celestial—celest
command—com
compartment-compt
condition—cond
construction—const
controls-cent
conversion—conver
cylinders—cyl

degrees—deg
diameter—dia
distribution—distr
doors—drs

ejection—eject
electrical—elec
electronic—elect
emergency—emer
enclosure-encl,enc
equipment—equip
exhaust-exh
extension—ext

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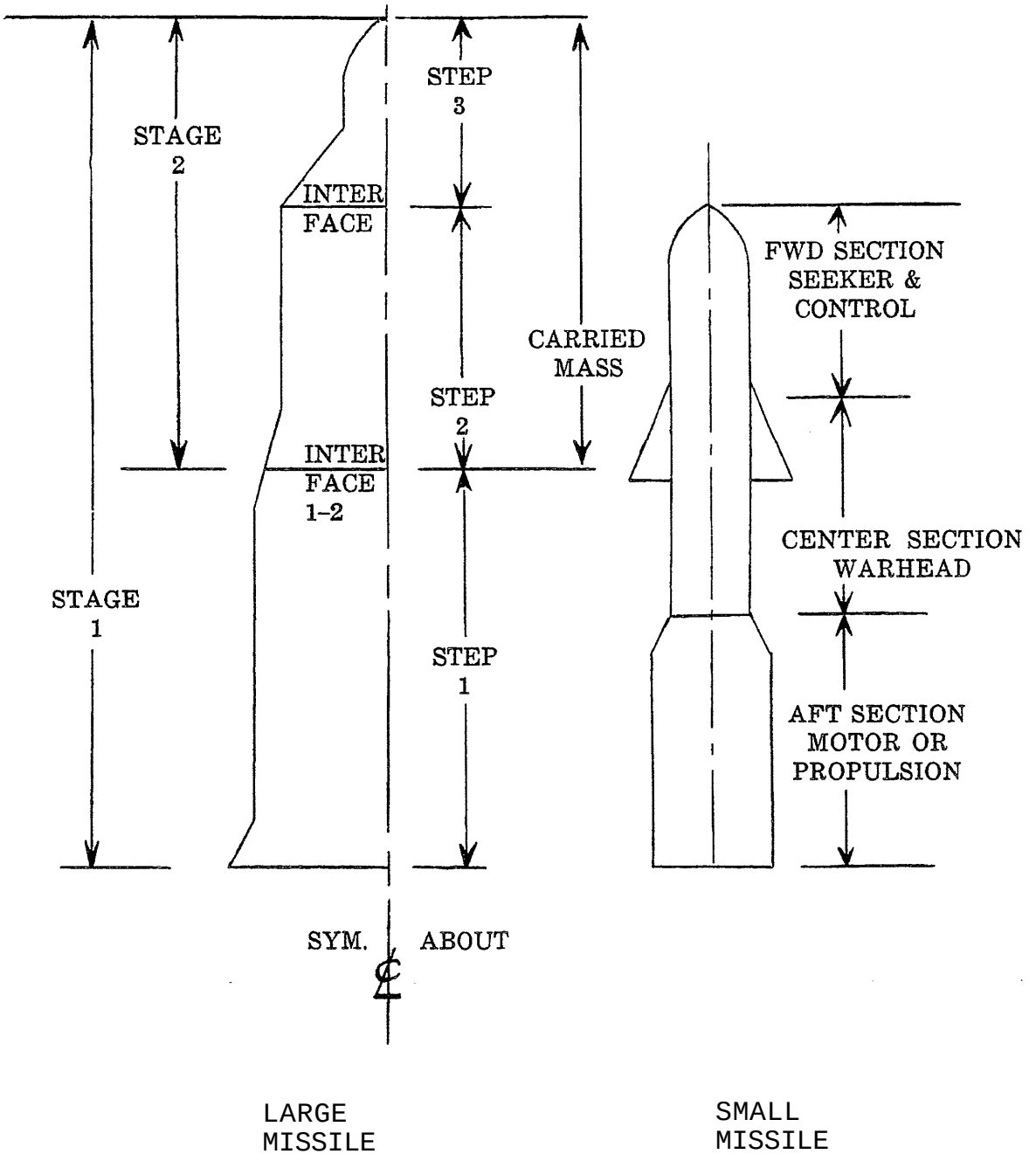


Figure 1

SAMPLE--MISSILE STEP-STAGE AND SECTIONS

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fastener–fastnr, fast
feet-ft
fitting-ftg
forward–fwd
fuselage-fus

gallons–gal
gear–gr
generator–gen
group-gr

hydraulic–hyd

inboard–inbd
inches–in
incidence-incid
include-incl
induction–ind
injection–inj
installation-instal
instrumentation–inst
integral–int
intermediate-inter

launching–launch
liquid–liq
location–loc
longeron–longn
longitudinal–long
lubricating–lub
Lubrication–lub

material–matl
rnaximum-max
mechanism–mech
miscellaneous–mist
missile-msl
mounts–mts

nacelle-nac
navigation–nav
number–no

operating–opr
outboard–outbd
oxidizer–oxidiz
oxygen–oxy

panels–pnl
parachute–para
pneumatic–pneu
power-pwr
pressur-press
propellers–prop
protection–prot

quantity-qty

release-rel
retraction–retr

secondary--sec, secondry
seconds-see
section–sect
stabilizing–stab
stiffeners–stiff
structure–str
structural–str
supports+sup
surface-sur
synchronizing–syn

temperature-temp
theoretical–theo

variable–var
ventilating–vent
volume-vol

weight-wt

4. GENERAL STATEMENTS AND REQUIREMENTS

4.1 Weight control systems shall provide for determination of breakdown weight data required by the statements, for consistency of allocation in accordance with the item allocations of the Detail Weight Statement, Part II, and the principles and intent outlined above in paragraph 1.3. Further structural and equipment breadowns, and generous use of footnotes, are encouraged when such practices will further understanding of the tabulated data and aid in the intended purposes.

4.2 The Detail Weight Statement lists entries which generally apply to all missiles. However, because of the varied missile con-

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figurations possible, some groupings do not list all required entries; hence, limited write-in spaces are provided for expansion of the listed groupings and for new installations. Insofar as practicable, nomenclature for write-ins shall identify the functions of the items, and allocation shall be made consistent with the principles outlined above in Section 1.3. Use of the words, "provision" and "installation," shall be avoided unless further breakdown is given to indicate what is included in the item. Supplementary sheets may be inserted when insufficient space is provided for new or unusual installations or expanded breakdowns.

4.3 All bolts, nuts, rivets, etc., and loose items of attachment shall be allocated to the same group as the item being attached.

4.4 Equipment and system supports shall be allocated to the same group as the items being supported. Multipurpose supports shall be allocated to the primary function. Where a primary allocation cannot be determined by observation, such as may occur for fairings, fairleads, supports for wire harness, brackets, etc. an arbitrary allocation may be made.

4.5 Where provisions are made in both weight empty and mission load categories for a given installation, allocation shall be made to the weight empty or mission load in accordance with the category allocation assigned in the model or detail specification or in the general design specifications and/or instructions of the procuring activity. If a conflict in category allocation of any item occurs between weight statement and model or detail specification, the latter shall govern in all cases.

4.6 All items of power distribution systems (electrical, hydraulic, and pneumatic), including fluid in the respective tubing for hydraulic systems, from main distribution points to actuating units shall be allocated with the items operated. All components from source of power to main distribution points shall be considered to comprise the main system. The distribution system for a given item shall be assumed to include all components which would be unnecessary and could be removed if power actuation of the item were to be eliminated. Where several successive branches from a distribution system are involved, and a clear cut main distribution point is not readily apparent, it shall be assumed that power is supplied to a given geographic location primarily for the function requiring the largest power, and to other functions in the sequence of descending power requirements.

4.7 When abbreviations are necessary for write-ins due to lack of space, the alleviations as required by Section 3 shall be used.

4.8 When business machine methods are utilized and reports are produced automat ically by machines, all pages as shown on the Summary and Detail Weight Statements (Parts I & II) must be identically reproduced. Where write-in items are necessary, or where an existing item is deleted or altered in any manner, a flag entry consisting of a double lozenge (◻◻) shall be made in the right hand margin directely opposite line.

4.9 Page size. Page sizes of 8½ x 11 inches for Parts I and II, and 11 x 22 inches (foldable to 8½ x 11 inches) for Part III, with uniform spacing of 6 lines per inch, shall be used for these weight statements and forms. Parts I, II and III reproduced herein are photographic reproduction of pages prepared in accord with the above sizes, and therefore represent standards of content and arrangement only. Full size copies of these forms, printed on vellum paper and suitable for ozalid, blueprint and similar methods of reproduction, may be obtained from the procuring activity or as directed by the contracting officer. Such requests should specify "Full size vellum forms Part I, II and/or III, or all, as applicable of MIL-STD-176."

4.10 All pages not used for the presentation of data may be omitted from the reports.

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List omitted pages on page 1 of Part I, or page 1 of Part II.

5. DETAIL REQUIREMENTS

5.1 Detail Instructions and Glossary of Terms (Specific items in the weight statements are identified by page-line numbers of the weight statement concerned).

5.1.1 Summary Weight Statement (Part I). The Summary Weight Statement gives a two-page breakdown of the missile weight empty as summarized from the Detail Weight Statement. This is followed by the Mission Load Breakdown to derive the launch-lift off gross weight. In addition, descriptive and design data of interest to weight engineers are included in the Summary Weight Statement.

5.1.1.1 Weight Empty (Pages 2 and 3). The steps and/or section subdivisions are to be indicated at the top of each page in the space provided and must coincide with those of the Detail Weight Statement. The group totals can be carried forward from the cor. responding group total listed at the bottom of the page in the Detail Weight Statement. Additional subtotal breakdowns are listed to show a more comprehensive summary, when warranted.

5.1.1.2 Mission Load (Page 4). The Mission Load is composed of all fixed and disposable items to be added to the weight empty to derive the maximum weight for launch (airborne missiles) or lift off (large surface launched missiles). All items jettisoned or disposed of during prelaunch and fright shall be deducted on the Weight Summary for Performance Analysis, Appendix II- to derive the weights at discrete points of the fright profile. Add supplemental page 4A if necessary for multi-section, multi-stage missiles.

5.1.1.3 Descriptive and Weight Criteria Design Data. These data include the majority of parameters used in the development of weight estimating techniques and comparative studies. In addition to entries for specific design criteria, various graphic plots are required to show deviations in the parameters for the complete fright duration. This will enable a review of the maximum values for a parameter as well as other discrete points of the flight profile.

Page 5—

Items 3 through 18 are self explanatory.
Items 19 through 26— Indicate the internal gage pressure that exists in each of the various tanks per design during the several operational phases noted. These are gas pressures and do not include propellant head pressures. The pressures are applicable to solid propellant tanks as well as liquid types. The vertical columns may be divided and headed by F & O to tabulate the fuel and oxidizer systems.
Item 27 through 33— Design Temperature.— Indicate by graphic plot in the vertical column spaces (or by an attachment) the skin temperature midlength of each tank and appropriate point on the nose cone vs. time of flight. Two graphs will be necessary for liquid systems, one each for fuel and oxidizer.
Item 34 through 44— Design Volumes.— The vertical column may be divided and headed by F & O to indicate the fuel and oxidizer systems. Gross Volume is the complete internal volume. Minimum ullage is the least amount of volume that will never be occupied by propellant in each of the various tanks. Nominal loadings may or may not be the same as maximum loaded volumes. Nominal minimum residuals is the volume in each of the items listed that is never available to the engines.
Item 48- Design Gross Weight is the gross weight of each step that is the basis for overall design.
item 49— Ultimate Safety Factor is the minimum ratio of the design load over the expected or applied load, where structural failure would occur if design load were realized.
Items 53 & 54— Indicate the most critical

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design load condition and the ultimate load factor for the critical condition.

Page 6–

Item 3 through 9– Indicate by graphic plot in the vertical column spaces (or by an attachment) the thrust and mass flow rate-M vs. time for the main and secondary engines for total time of burning–for the nominal design condition.
Item 10, 11 and 12– For the main and secondary engines, indicate the specific impulse (ISP) in lb-see/lb. (sec.) and state whether based on sea level or vacuum environment.
Item 13 through 17- Indicate by graphic plot the velocity-V, longitudinal load factor-N_{x9} dynamic pressure-q $(\frac{1}{2}V^2)_p$ range-R, altitude-A and pitching angular acceleration vs. time. Units for these values will be determined by the reporting activity. The graphs should be clear and concise. They may be arranged in the vertical column if space permits or assembled on a supplemental page.
Item 19– Propellant Step Ratio is the weight of usable propellant in the step divided by the gross weight of the step, per design.
Item 20– Stage Mass Ratio for each stage is the stage design weight at the beginning of effective burning divided by the stage weight at the end.
Item 21– Impulse to Step Weight Ratio for each step is the impulse (Ib-see) produced by the usable propellant in the step divided by the gross weight of the step, per design.
Item 22- Impulse to Stage Weight Ratio for each stage is the impulse (lb-sec) produced by the usable propellant in the active step divided by the design gross weight of the stage.
Item 23– Ultimate Payload Ratio for the overall vehicle is the weight of ultimate payload divided by the design gross weight of the first stage. Examples of ultimate payload are warhead and fuse, nosecone, instruments and cargo.
Item 24- Stage Payload Ratio for each stage is the payload weight of the stage divided by the stage design weight at the beginning of effective burning, Typically, payload is the carried weight of a stage above the active step. (Note: The final payload ratio of the overall vehicle is the combined product of a11 the stage payload ratios.)
Item 25, 26, and 27– Indicate the ultimate gage pressure to which the accumulator is designed and give the specific number designation for the fluid.
Item 28 and 29– Indicate the ultimate gage pressure to which the accumulator is designed and indicate whether air or some other gas.
Item 31 through 34, and 39 through 42- Indicate output power for generators and input power for inverters and converters.
Item 44 and 45– Indicate the design capacities for heating and cooling.
Item 47 and 48– This is applicable to roll stabilized vehicles only.
5.1.2 Detail Weight Statement (Part II). The Detail Weight Statement gives a 21-page summary of the many detail parts and assemblies that make up the weight empty of the guided missile. It provides for the collection of detail weight data of diverse missiles into a standardized functional summation and at the same time provides identifiable geographic location data. The weight empty is subdivided into major groups on a functional basis–that is, body, takeoff and recovery, propulsion, guidance, etc. These are Detail Weight Statement page titles which are carried forward to the Summary Statement. The identifiable geographic breakdown is given in the six columns on the right of each page, and each column is identified by step number, section limits, etc.

5.1.2.1 Items serving multi-purpose functions. These items which cannot be isolated as serving a single function, and which are integral with the structure of aerodynamic surfaces or body, shall be allocated to the appropriate structural group by write-in entries using functional identifying nomenclature. Typical in this category are:

(a) Ducting for air supply, bleed air and exhaust in which skin or webs serve as part of the duct contour. The duct wall and stiffeners which can be isolated pri-

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marily as duct system are allocated to "Propulsion Group—Air Breathing-Air Induction System or Exhaust System" or to "Environmental Protection Group—Temperature Control-Structure, Ducts."

(b) Integral tanks in which skin, webs, solid bulkheads and load carrying seal fittings are utilized to form the cavity. The separate tank supports, sealant material, nonload carrying fittings are allocated to "Propulsion Group-Propellant Systems."

(c) Skin, ablative material or fabricated special material required for adverse environments including elevated temperature. Include these in structural groups. The nonstructural temp-control systems for structure or equipment are allocated to "Environmental Protection Group."

(d) Those radomes which do not form normal contour of aerodynamic surface or body. These are allocated to the Guidance or Electronics Groups.

(f) Those removable structural panels or structural doors for special functions, These are allocated to the basic structural groups.

5.1.2.2 Items common to the several structural groups.

(a) Joints, Splices and Fasteners. These items include all doublers, fillers, spacers, gussets, angles, clips, rivets, bolts, nuts, screws that should insofar as practicable deallocated by structural function served.

(b) Bulkheads, Frames, Cover and Lengerons. Bulkheads are the main structural members which assist in carrying loads around major cutouts, or which distribute major internal or external concentrated loads and may also act as partitions to separate an area into physical compartments. Body bulkheads should be listed individually and identified by location. Frames are similar to bulkheads but usually of relatively lighter construction and serve the function primarily of stiffening the cover or skin to maintain contour under load. Longerons serve to distribute the load longitudinally along with longitudinal partitions which contribute to body strength.

(c) Exterior Finish. This includes paint and finish for protection from the elements. Any interior finish is coded to the item it covers.

5.1.2.3 Aerodynamic Surfaces (Page 2 and 3). This group is used for all primary lifting and aerodynamic control surfaces, both fixed and movable.

5.1.2.3.1 Page 2. Identify the various wings such as "vertical," "horizontal," etc. Auxiliary control surfaces which are attached to the wings shall be listed on page 3--"Surfaces."

(a) Leading Edge (item 16, 41) is defined as all structure forward of the front spar.

(b) Trailing Edge (Item 17,42) is defined as all structure aft of the rear spar.

(c) Core (Item 12, 37) is defined as sandwich hexcell (alum., stl., etc) or foam.

(d) Exterior finish is to be allocated to covering,

(e) Insulation in the wing, fin or stabilizer shall be a "write-in" item.

(f) Wing-body splices and fittings are to be allocated to the wing.

(g) Identify aero surface heat shields.

5.1.2.3.2 Page 3. The surfaces on this page are to be identified and labeled whether "fixed" or "moveable." Also identify aerodynamic surface heat shields.

5.1.2.4 Body Group (Pages 4 and 5). The body group is arranged so that separate seetions may be broken down within each step column for multi-stage vehicles. Where total section weights as assembled are desired by the procuring agency and the above body format is used, addition sheets may be added to other groups where applicable to show

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breakdowns by section. The body pages are broken up into distinct structural divisions compatible for booster or aerodynamic missiles. Identify sections which are integral fuel tanks by suitable title for example "forward fuel tank." Identify by write-in or note that structure which is jettisoned during flight boost phase. Write in, where applicable, the items used to provide the vehicle with the flexibility to operate in adverse environment; for example, (a) ablative material or other protection for high temperature environment, (b) integral structural provisions for liquid cooling.

5.1.2.5 Takeoff and Recovery (Page 6). This group is intended to contain items which are primarily used for missile launching and recovery. Enter descriptive or location data, as appropriate, for clarification of the function served.

5.1.2.6 Propulsion Group (Pages 7, 8, 9, 10, and 11). This group covers those items whose primary function is to provide flight path thrust or acceleration. Other propulsion devices, such as those used for electrical power or to maintain control of orientation are coded elsewhere. (Except for thrust vectoring systems which are integral with the engine and these must be allocated to the engine.)

5.1.2.6.1 Liquid Rocket (Page 7).

(a) Propellant Feed System (Item 11) –This system is only that part of the total propellant feed system that is attached to, immediately adjacent to or integral with the engine installation.

(b) Engine Mounts (Item 26)–The thrust structure directly supporting the engine is coded here.

(c) Controls and Valves (Item 32)– Include here any controls or valves which are associated with the operation of the engine, usually as supplied by the engine manufacturer for the start and shutdown of the system.

(d) Propellant Control System *(Item 45).* –List here any control system such as gaging systems for propellant loading, propellant utilization systems which sense propellant tank levels and send appropriate control signals to the engine and liquid level sensors. These devices will usually be provided by the propellant tank manufacturer, thus helping to differentiate them from Item 32.

5.1.2.6.2 Solid Rocket (Page 8). This page covers the complete solid propellant rocket propulsion system and all items associated therewith. A1l other propulsion pages may be deleted when a solid system is used for the main propulsion unit. Unexpended or inert propellant is not included here but is provided for in the Mission Load.

(a) Insulation–Internal (Item 32) List the total weight including that which is burned.

5.1.2.6.3. Air Breathing and Nuclear (Page 9).

(a) Air Breathing (Item 4)–Code all air breathing systems here. Write in nacelle, pylon, engine section, etc. with adequate structural breakdown (skin, bulkheads, longerons, etc.) provided that a clear distinction (i.e., not integral with the body) can be made.

(b) Shielding–Thermal (Item 32). include all shielding needed for reactor functioning. Shielding for environmental protection is coded to page 19.

5.1.2.6.4 Propellant Systems. The propellant systems are (Page 10 & 11) composed of those parts that contain and feed to the propulsion unit the propellants needed. As a minor deviation in the normal functional subdivision of vehicles, lubrication systems are included with propellant systems. The propellant systems are presented on two pages, one for the fuel system and one for the oxidizing system. The fuel system page only should be used for a monopropellant

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system. Identify all nonintegral tanks. Integral tanks are coded to body and cross-referenced here.

5.1.2.7 Power Generating Group (Page 12, 13, & 14). This group provides for those systems used to generate power for purposes other than propulsion. The source of initial power is included under prime power source; however, once power has been generated, any further conversion equipment is listed under its specific system such as electric or hydraulic.

(a) Prime Power Source

(Page 12)—Indicate if the prime power source includes an integral pump or generator; all separate pumps or generators are to be included in the distribution system (i.e., hydraulic—pneumatic or electrical).

5.1.2.8 Orientation Controls Group (Page 15 and 16). This group provides two pages for the two conceivable means of orientation control, i.e., Aerodynamic and Thrust-Reaction. Write in for each group the type power as indicated at the bottom of the page.

5.1.2.9 Guidance Group (Page 17). The guidance group is divided into three major subgroups and, as an aid in understanding, these subgroups can be likened functionally to parts of the human body as follows:

Guidance Source . . . to the eyes and other sense organs;

Prime Reference . . . to the brain and organs of equilibrium; and the

OutPut System . . . to the nervous system.

5.1.2.10 Electronics Group (Page 18). This group provides for electronic equipment riot associated with guidance of the missile. It will include communication equipment, beacons and navigational aids. Identify by note or write-in that equipment which serves to indicate malfunctioning.

5.1.2.11 Environmental Protection Group (Page 19). This group provides for those systems which regulate the environment to protect the equipment and structure in order that it may perform its function. Insulation, not integral with structure, is to be allocated to the functional group requiring it. Self-contained liquids required for this group are to be listed in Items 12 and 26. Cooling fluid disposed of during flight is to be allocated to Mission Load.

5.1.2.12 Armament Group (Page 20). The Armament Group provides for the allocation of items functioning as supporting devices to the primary armament listed in the Mission Load. The items allocated to this group are not disposed of prior to impact.

5.1.2.13 Miscellaneous Group (Page 21). This group provides for all the other equipment serving minor functions which are not provided for in other groups. "Expendable" visual identification items such as flares are to be allocated to Mission Load.

5.1.2.14 Appendix I to Detail Weight Statement (Part II) Structural Increments for Design Features—Page i of the appendix provides for listing the net structural weight increments for special functions, installations and other design features. Insofar as practicable, the net increment buried or masked in "basic structural" and quantitatively evaluated by some rational procedure should be listed for such items as body joints, major cutouts, removable structural panels, etc. Page ii of the appendix provides for amplifying and clarifying remarks pertaining to the increments listed on page i.

5.1.3 Weight and Balance Status (Part IIIA and IIIB.).

(a) Part IIIA Page (1) is used to present the weight and balance status of the same groups as shown in the group summary statement for a single stage missile. The group headings are considered to be self explanatory. The propellant ratio is also entered at the bottom of the page for the status conditions indicated by the column headings.

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(b) Part IIIA Page (2) provides for listing separately each major change of Government responsibility. The weight variations in Government Furnished Equipment (GFE) is listed in column 12.

(c) Part IIIA Page (3) is used for major contractor changes. Each change is to be identified with a brief description and by a significant change number. The miscellaneous small contractor changes may be all grouped into column 12.

(d) Part IIIB Page (1) intended for multistage missiles, lists the current weight of the groups of the summary group statement and provides a concise form for comparison of the current weight with the specification weight. The weights are composed of two subtotals: Weight Empty and Mission Load. These are totaled to derive the overall liftoff weight. The columns across the page are used to show all of the totals by steps and a total of all steps. A revised specification weight is computed for each subtotal and total vehicle by adding the current government changes to the previously current specification weight. The resulting revised specification is compared with the current design weight to compute a current over/under weight. All of these computed totals are listed in the applicable step column. The current center of gravity and the propellant ratio of each step are shown at the bottom of the form. These terms, "stage" and "step," are as defined in paragraph three. The terms before and after separation, as used in determining C. G. configurations, are defined as follows:

(1) After separation is the fully loaded stage just after liftoff or just after an upper stage separates from its lower step.

(2) Before separation is the stage just before its lowest step separates.

(e) Part IIIB-Page 2 (No form shown). This page is used to list the changes since the last status. The major changes are listed separately and miscellaneous minor changes may be grouped together. Both of these type changes; however, are to be tabulated by group, responsibility and step. Each major change is to be identified with a brief description and by a significant change number. The change listing is to be totaled so that the current group weights on page 1 can be reconciled with the previous status. Significant changes to the C. G. and propellant ratio are also to be explained on this page.

6. WEIGHT SUMMARY FOR PERFORMANCE ANALYSIS—APPENDIX II.

This appendix is intended as a general guide for the preparation of a Weight Summary for Performance Analysis as may be required by the procuring agency.

(a) Figure 2 exemplifies the type of tabulation desired to provide the weight vs. time history at discreet points during the flight trajectory of the vehicle.

(b) Because of the large number of possible configurations, only certain general items have been listed. Space for write-ins has been provided to allow for the listing of items pertaining to a specific missile.

(c) The tolerance to be listed (when required) is that tolerance that has previously been determined from other requirements established by the procuring agency.

(d) A list of terms and definitions most widely used is furnished to promote the use of common terminology when making use of Figure 2 or its equivalent.

Term	Definition
Effective Burning	The propellant or time used for useful acceleration of the vehicle. Start of effective burning begins either at liftoff, or separation from a previous Step when ignition occurs prior to separation, or at a specified

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Term	Definition
	thrust level during thrust buildup whenever ignition follows separation.
End of Effective Burning	This occurs when aspecified thrust level is reached during thrust decay.
Useful Propellant	That mass of propellant consumed during the programmed nominal, effective burning time.
Usable Propellant Reserves	That mass of usable propellant which under expected flight conditions is not normally utilized or expended during effective burning time.
Maximum Usable Propellant	Useful Propellant mass plus the Usable Propellant Reserves.
Stage Mass at beginning of Effective Burning	The stage mass at the instant the thrust has reached a prescribed level, or after a specified delay subsequent to reaching this level, during the thrust buildup of the Step.
Stage Mass at End of Effective Burning	The stage mass at the instant-the thrust reaches a prescribed level during thrust decay of the step.
Minimum Stage Mass at End of Effective Burning	The stage mass at the instant the thrust level reaches a prescribed level during thrust decay of the Step. This assumes usage of the Maximum Usable Propellant.
Step Mass at End of Action Time	The least mas of the Step under normal conditions. The Step has deleted its Maximum Usable Propellant and its Expended Inerts.
Minimum Step Mass at End of Action Time	The minimum mass the Step will have under Alternate conditions. The Step has deleted its Maximum Usable Propellant and its Expended Inerts.
Pre-Effective Burning Expended Inerts	That mass of inert material ablated, eroded, jettisoned or in any way removed from the vehicle prior to the beginning of Effective Burning Time of the Step. This includes propellant expended during thrust during thrust buildup.
Effective Burning Time Internally Expended Inerts	That mass of inert material expended through the nozzle during the Effective Burning Time of the Step.
Post-Effective Burning Expended Inerts	That mass of inert material ablated, eroded, jettisoned or in any way removed from the vehicle after Effective Burning Time of the Step and prior to separation of the Step.

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APPENDIX II
WEIGHT SUMMARY FOR PERFORMANCE ANALYSIS

Missile No.:_____

Item	Step Weight Summary (lb)			Stage Weight Summary (lb)
	Step I	Step II	Step III	
Aero. Surfaces				
Body				
Takeoff & Recovery				
Propulsion				
Power Generating				
Orientation Control				
Guidance				
Electronics				
Environmental Protection				
Armament				
Miscellaneous				
WEIGHT EMPTY				
Payload				
Telemetering				
DRY WEIGHT				
Tolerance Step I —				
Tolerance Step II —				
Tolerance Step III —				
Gas				
Lubricant				
Propellant Loaded				
Liquid Propellants				
Solid Propellants				
LOADED WEIGHT				
Start and Ground Losses				
Liquid Propellant				
LIFTOFF WEIGHT—STAGE I				

Figure 2

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APPENDIX II
WEIGHT SUMMARY FOR PERFORMANCE ANALYSIS

Missile No.:_____

Item	Step Weight Summary (lb)			Stage Weight Summary (lb)
	Step I	Step II	Step III	
Propellants Consumed during Stage I Flight WEIGHT AT STAGE I BURNOUT				
Step I Burnout Weight Start Losses STAGE II START WEIGHT				
Decoy Fairings Propellant Consumed during Stage II Flight WEIGHT AT STAGE II BURNOUT				
Propellant Consumed during Vernier Flight STAGE II WEIGHT AT VERNIER BURNOUT				
Step II Weight at Vernier Burnout STAGE III WEIGHT AT SEPARATION				

Figure 2
(con.)

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NAME . . . . . PAGE ----------

DATE . . . . . MODEL ----------

REPORT ----------

MIL-STD-176A PART I

SUMMARY WEIGHT STATEMENT
GUIDED MISSILE AND SPACE LAUNCH VEHICLES

ESTIMATED-CALCULATED-ACTUAL

(Cross Out Words Not Applicable)

List Pages Omitted ----------

APPROVED NAME ----------

TYPE ----------

MANUFACTURED BY ----------

CONTRACT NUMBER ----------

CONTRACTOR'S DESIGNATION ----------

ROUND OR SERIAL NUMBER ----------

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PAGE_____

NAME_____ SUMMARY WEIGHT STATEMENT MODEL_____

DATE_____ Weight Empty REPORT_____

1	STEP						
2	SECTION						
3							
4	AERO. SURFACE GROUP						
5	WING						
6							
7							
8							
9	FIN						
10							
11							
12	STABILIZER						
13							
14							
15							
16	BODY GROUP						
17							
18							
19							
20							
21							
22							
23							
24	TAKEOFF & RECOVERY GROUP						
25							
26							
27							
28	PROPULSION GROUP						
29	LIQUID SYSTEM						
30	ENGINE						
31	ENGINE MOUNT						
32	PROPELLANT FEED SYS.						
33	CONTROL & VALVES						
34	STARTING SYS.						
35	PROPELLANT CONTROL SYS.						
36							
37							
38							
39	SOLID SYSTEM						
40	NOZZLE SYSTEM						
41	PROPELLANT CASE						
42	THRUST CONTROL						
43	INSULATION						
44	IGNITER, SAFE & ARM						
45							
46	AIR BREATHING SYS.						
47							
48	NUCLEAR SYS.						
49							
50	FUEL & LUB. SYS.						
51							
52	OXIDIZER & LUB. SYS.						
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL THIS PAGE						

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PAGE_____

NAME_____ SUMMARY WEIGHT STATEMENT MODEL_____

DATE_____ Weight Empty REPORT_____

1	STEP						
2	SECTION						
3							
4	POWER GENERATING GROUP						
5	PRIME POWER SOURCE						
6	HYDRAULIC & PNEUMATIC						
7	ELECTRICAL						
8							
9							
10							
11	ORIENTATION CONT. GROUP						
12	AERODYNAMIC						
13	THRUST & REACTION						
14							
15							
16							
17	GUIDANCE GROUP						
18							
19							
20							
21							
22							
23	ELECTRONICS GROUP						
24							
25							
26							
27							
28							
29	ENVIRON. PROT. GROUP						
30	TEMP. CONT. — STRUCT.						
31							
32	EQUIP.						
33	COMP. PRESS. SYS.						
34	GROUND SERVICE PROV.						
35							
36							
37	ARMAMENT						
38							
39							
40							
41							
42							
43	MISCELLANEOUS						
44	STEP SEPARATION SYSTEM						
45	DESTRUCT SYSTEM						
46	EMERGENCY EQUIPMENT						
47	VISUAL IDENTIFICATION						
48							
49							
50							
51							
52							
53							
54							
55	COLUMN TOTALS						
56	CARRIED FORWARD FROM PAGE 2						
57	TOTAL—WEIGHT EMPTY						

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PAGE_____

NAME_____ **SUMMARY WEIGHT STATEMENT** MODEL_____

DATE_____ **Mission Load and Launch Weight** REPORT_____

1	STEP						
2	SECTION						
3							
4	PROPELLANT—USABLE.						
5	LIQUID FUEL						
6	OXIDIZER						
7							
8	SOLID						
9							
10	SECONDARY POWER						
11	STEP SEPARATION						
12	LANDING ENGINE						
13	RETRO SYSTEM						
14	REACTION CONTROLS						
15							
16							
17	PROPELLANT—UNUSABLE						
18	TRAPPED						
19							
20	BIAS OR OUTAGE						
21							
22	BOILOFF RETAINED						
23	INERT						
24	LUBRICANT						
25	PRESSURE MEDIUM						
26							
27							
28							
29							
30							
31							
32	TEST EQUPIMENT						
33	TELEMETERING						
34							
35							
36	BALLAST						
37							
38	COOLING LOSSES						
39	STRUCTURE						
40	FLUIDS						
41	SOLIDS						
42	EQUIPMENT						
43	CARGO						
44							
45	SEPARATION ROCKET CASES						
46							
47	PAYLOAD						
48	CARGO						
49	NOSE CONE OR.						
50	RE-ENTRY VEHICLE						
51	WARHEAD						
52							
53							
54	COLUMN TOTAL MISSION LOAD						
55	TOTAL SECTION WT. M.L.+W.E.						
56	STEP OR STAGE WEIGHTS						
57	TOTAL—LAUNCH OR LIFT OFF WEIGHT						

ADD SUPPLEMENTAL PAGE 4A IF NECESSARY

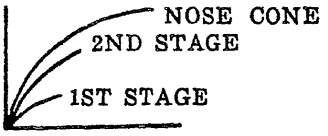
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NAME_____ PAGE_____

DATE_____ MODEL_____

REPORT_____

SUMMARY WEIGHT STATEMENT
Descriptive & Weight Criteria
Design Data

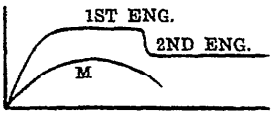
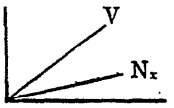
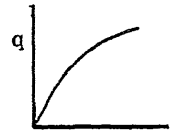
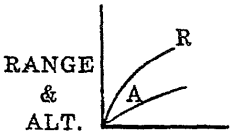
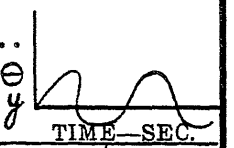
1	STEP					
2	SECTION					
3	AERO SURFACE GROUP					
4	SURFACE IDENTIFICATION					
5	QUANTITY					
6	GROSS AREA SQ. FT./MSL					
7	NET AREA SQ. FT./MSL					
8	CONSTRUCTION & MATL.					
9	BODY GROUP					
10	MAX. LENGTH (IN.)					
11	MAX. WIDTH (IN.)					
12	MAX. DEPTH (IN.)					
13	VOLUME (CU. FT.)					
14	WETTED AREA (SQ. FT.)					
15	CONSTRUCTION & MATL.					
16	PROPELLANT NOMINAL DENSITY					
17	FUEL—LB./CU. IN.					
18	OXIDIZER—LB./CU. IN.					
19	FUEL & OXIDIZ. TANKAGE DATA					
20	DES. INT. PRESS.—ULT. PSI					
21	HANDLING					
22	LOADING					
23	GROUND LAUNCH					
24	MAX. Q					
25	SPACE LAUNCH					
26	FLIGHT TERMINATION					
27	DES. TEMP.—DEG. FAHR.					
28						
29						
30						
31						
32						
33						
34	DES. VOLUMES—CU. FT.					
35	GROSS					
36	MIN. ULLAGE					
37	NOMINAL LOADING					
38	MAIN TANKS					
39	MAIN PLUMB. & ENGS.					
40	OTHER TANKS & PLUMB.					
41	NOM. MIN. RESIDUALS					
42	MAIN TANKS					
43	MAIN PLUMB. & ENGS.					
44	OTHER TANKS & PLUMB.					
45						
46						
47						
48	DESIGN GROSS WEIGHT					
49	ULTIMATE SAFETY FACTOR					
50	ULT. VERT L. F. AT D.G.W.					
51	MAX. ALLOW. WIND VEL.					
52	AT GROUND LAUNCH—KNOTS					
53	CRITICAL DESIGN LOAD COND.					
54	ULT. L.F. FOR CRITICAL LOAD					
55	COLUMN TOTALS					
56						
57	TOTAL					

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NAME_____ DATE_____

SUMMARY WEIGHT STATEMENT
Descriptive & Weight Criteria
Design Data—(con.)

PAGE_____ MODEL_____ REPORT_____

1	STEP						
2	SECTION						
3	MAIN & SECONDARY ENG. DATA						
4	NOMINAL DESIGN CONTD.						
5							
6	THRUST—						
7	LBS.						
8	MASS FLOW						
9	RATE (LBS./SEC.)						
10	SPECIFIC INPULSE S.L. OR VAC.						
11	MAIN ENG. LB. SEC./LB.						
12	SEC. ENG. LB. SEC./LB.						
13	VELOCITY						
14							
15	*N_x						
16							
17							
18	RATIOS						
19	PROPELLANT STEP TO G.W.						
20	STAGE MASS.						
21	IMPULSE TO STEP WEIGHT						
22	IMPULSE TO STAGE WEIGHT						
23	ULTIMATE PAYLOAD						
24	STAGE PAYLOAD						
25	HYDRAULIC SYSTEM						
26	DESIGN PRESS—ULT. PSI						
27	FLUID TYPE						
28	PNEUMATIC OR GAS SYSTEM						
29	DESIGN PRESS.—ULT. PSIG						
30	AC ELECTRICAL SYSTEM						
31	GEN. POWER-VOLT AMPS.						
32	GEN. POWER-VOLTAGE						
33	INVER. PWR-VOLT AMPS.						
34	INVER. PWR-VOLTAGE						
35	DC ELECTRICAL SYSTEM						
36	BATTERY—TYPE						
37	BATTERY—CAP. K.W. HRS.						
38	BATTERY—VOLTAGE						
39	GEN. POWER—K.W.						
40	GEN. POWER—VOLTAGE						
41	CONVT. PWR—K.W.						
42	CONVT. PWR—VOLTAGE						
43	ENVIRONMENTAL CONT. SYS.						
44	HEATING CAP.—EQUIP-BTU/HR						
45	COOLING CAP.—EQUIP-BTU/HR						
46							
47	MAX. ROLL VEL.—RAD/SEC.						
48	MAX. ROLL ACCEL—RAD/SEC. SQ.						
49							
50	GUIDANCE—TYPE						
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL						

*LONGITUDINAL LOAD FACTOR

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NAME . . . . . PAGE----------
DATE . . . . . MODEL_____
REPORT_____

MIL-STD-176A PART II

DETAIL WEIGHT STATEMENT

GUIDED MISSILE AND SPACE LAUNCH VEHICLES WEIGHT EMPTY

ESTIMATED-CALCULATED-ACTUAL
(Cross Out Words Not Applicable)

List pages omitted _____

APPROVED NAME_____

TYPE_____

MANUFACTURED BY_____

CONTRACT NUMBER_____

CONTRACTOR'S DESIGNATOR'S_____

ROUND OR SERIAL NUMBER _____

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PAGE_____

NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Aerodynamic Surfaces** REPORT_____

1	STEP						
2	SECTION						
3							
4	WING						
5	BASIC STRUCTURE						
6	COVERING						
7	SPAR CAPS						
8	SPAR WEBS & STIFF.						
9	RIBS						
10	STIFF.-SPANWISE						
11	STIFF.-CHORDWISE						
12	CORE						
13	JOINTS, SPLICES & FAST.						
14							
15							
16	LEADING EDGE						
17	TRAILING EDGE						
18	SECONDARY STRUCTURE						
19	ACCESS DOORS & PANELS						
20	FAIRING & FILLETS						
21	TIPS						
22							
23	HINGES & SHAFTS						
24							
25							
26							
27							
28							
29	FINS OR STABILIZER						
30	BASIC STRUCTURE						
31	COVERING						
32	SPAR CAPS						
33	SPAR WEBS & STIFF.						
34	RIBS						
35	STIFF.—SPANWISE						
36	STIFF.—CHORDWISE						
37	CORE						
38	JOINTS-SPLICES & FAST.						
39							
40							
41	LEADING EDGE						
42	TRAILING EDGE						
43	SECONDARY STRUCTURE						
44	ACCESS DOORS & PANELS						
45	FAIRING & FILLETS						
46	TIPS						
47							
48	HINGES & SHAFTS						
49							
50							
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—THIS PAGE						

M INDICATES MOVABLE

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PAGE_____

NAME_____ MODEL_____

DATE_____ REPORT_____

DETAIL WEIGHT STATEMENT
Aerodynamic Surfaces

1	STEP						
2	SECTION						
3							
4	SURFACE*						
5	BASIC STRUCTURE						
6	COVERING						
7	SPAR CAPS						
8	SPAR WEBS & STIFF.						
9	RIBS						
10	STIFF.—SPANWISE						
11	STIFF.—CHORDWISE						
12	CORE						
13	JOINTS, SPLICES & FAST.						
14							
15							
16	LEADING EDGE						
17	TRAILING EDGE						
18	SECONDARY STRUCTURE						
19	ACCESS DOORS & PANELS						
20	FAIRING & FILLETS						
21	TIPS						
22							
23	HINGES & SHAFTS						
24							
25							
26							
27							
28							
29	SURFACE*						
30	BASIC STRUCTURE						
31	COVERING						
32	SPAR CAPS						
33	SPAR WEBS & STIFF.						
34	RIBS						
35	STIFF.—SPANWISE						
36	STIFF.—CHORDWISE						
37	CORE						
38	JOINT, SPLICES & FAST.						
39							
40							
41	LEADING EDGE						
42	TRAILING EDGE						
43	SECONDARY STRUCTURE						
44	ACCESS DOORS & PANELS						
45	FAIRING & FILLETS						
46	TIPS						
47							
48	HINGES & SHAFTS						
49							
50							
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—AERODYNAMIC SURFACES						

* IDENTIFY TYPE OF SURFACE M INDICATES MOVABLE

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PAGE_____

NAME_____ DETAIL WEIGHT STATEMENT MODEL_____

DATE_____ Body REPORT_____

1	STEP						
2	SECTION*						
3	BULKHEADS						
4							
5							
6							
7	FRAMES						
8	JOINTS, SPLICES & FAST.						
9	COVER						
10	LONGERONS						
11	UPPER						
12	LOWER						
13	LONGITUDINAL PARTITION						
14	FIREWALLS						
15	FLOORING & SUPPORTS						
16	INSULATION						
17							
18	FITTINGS						
19							
20	EXTERIOR FINISH						
21							
22							
23							
24							
25							
26	SECTION*						
27	BULKHEADS						
28							
29							
30							
31	FRAMES						
32	JOINTS, SPLICES & FAST.						
33	COVER						
34	LONGERONS						
35	UPPER						
36	LOWER						
37	LONGITUDINAL PARTITION						
38	FIREWALLS						
39	FLOORINGS & SUPPORTS						
40	INSULATION						
41							
42	FITTINGS						
43							
44	EXTERIOR FINISH						
45							
46							
47							
48							
49							
50	FITTINGS*						
51	HANDLING						
52	LAUNCH						
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL						

* IDENTIFY SECTION

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PAGE_____

NAME_____ DETAIL WEIGHT STATEMENT MODEL_____

DATE_____ Body REPORT_____

1	STEP						
2	SECTION*						
3	BULKHEADS						
4							
5							
6							
7	FRAMES						
8	JOINTS, SPLICES & FAST.						
9	COVER						
10	LONGERONS						
11	UPPER						
12	LOWER						
13	LONGITUDINAL PARTITION						
14	FIREWALLS						
15	FLOORING & SUPPORTS						
16	INSULATION						
17							
18	FITTINGS						
19							
20	EXTERIOR FINISH						
21							
22							
23							
24							
25							
26	SECTION*						
27	BULKHEADS						
28							
29							
30							
31	FRAMES						
32	JOINTS, SPLICES & FAST.						
33	COVER						
34	LONGERONS						
35	UPPER						
36	LOWER						
37	LONGITUDINAL PARTITION						
38	FIREWALLS						
39	FLOORING & SUPPORTS						
40	INSULATION						
41							
42	FITTINGS						
43							
44	EXTERIOR FINISH						
45							
46							
47							
48							
49							
50	FITTINGS*						
51	HANDLING						
52	LAUNCH						
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL						

* IDENTIFY SECTION

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Takeoff and Recovery** REPORT_____

1	STEP						
2	SECTION						
3							
4	TAKEOFF GEAR						
5	ATTACHMENT FTG.						
6							
7	TIE-DOWN FITTINGS						
8							
9							
10							
11							
12	DECELERATION SYSTEM						
13	RETRO SYSTEM						
14							
15							
16							
17							
18							
19	PARACHUTE SYSTEM						
20							
21							
22							
23							
24							
25							
26							
27	DRAG BRAKES						
28							
29							
30							
31							
32							
33							
34	IMPACT SYSTEM						
35	MAIN						
36							
37							
38							
39							
40							
41	AUXILIARY						
42							
43							
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54							
55	COLUMN TOTALS						
56							
57	TOTAL—TAKEOFF & RECOVERY						

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NAME_____

DATE_____

DETAIL WEIGHT STATEMENT

Propulsion Group-Engine Instal.

Liquid Rocket

MODEL_____

REPORT_____

1 STEP						
2 SECTION						
3 THRUST CHAMBER ASSEMBLY						
4 MANIFOLD						
5 INJECTOR						
6 COMBUSTION CHAMBER						
7 ACTUATOR MOUNT ATTACH.						
8 THRUST DOME OR TRAN. CONE						
9						
10						
11 PROPELLANT FEED SYSTEM						
12 TURBO PUMP FEED						
13 PUMP DRIVE ASSEM.						
14 EXHAUST SYSTEM						
15 PROPELLANT FEED LINES						
16 PROPELLANT DRAIN & VENT						
17						
18						
19 PRESSURIZED FEED						
20 CONTROLS						
21						
22 PROPELLANT FEED LINES						
23 PROPELLANT DRAIN & VENT						
24						
25						
26 ENGINE MOUNT						
27 THRUST STRUCTURE						
28 GIMBAL/SWIVEL						
29						
30						
31						
32 CONTROL & VALVES						
33 HYDRAULIC CONTROLS						
34 PNEUMATIC CONTROLS						
35 ELECTRICAL CONTROLS						
36 VALVES						
37						
38						
39 STARTING SYSTEM						
40						
41						
42						
43						
44						
45 PROPELLANT CONTROL SYSTEM						
46						
47						
48						
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53						
54						
55 COLUMN TOTALS						
56						
57 TOTAL—ENGINE INSTALLATION—LIQUID						

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PAGE_____

NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Propulsion Group-Engine Instal.—(con.)** REPORT_____

Solid Rocket

1	STEP						
2	SECTION						
3	CASE STRUCTURE						
4	FORWARD CLOSURE						
5	CENTER SECTION						
6	AFT CLOSURE						
7	ATTACH. FLANGES						
8	SKIRT. & ATTACH.						
9							
10							
11	NON-MOVABLE NOZZLE SYSTEM						
12	BLAST TUBE						
13	BLAST TUBE ATTACH. PROV.						
14	EXIT CONE						
15							
16							
17	MOVABLE NOZZLE SYSTEM						
18	BLAST TUBE						
19	BLAST TUBE ATTACH. PROV.						
20	HINGE MECH.—FIXED						
21	EXIT CONE						
22	HINGE MECH.—MOVABLE						
23	SEALS						
24							
25							
26	THRUST TERMINATION						
27							
28							
29							
30	THRUST REVERSAL						
31							
32	INSULATION—INTERNAL						
33	CASE						
34	NOZZLES						
35							
36							
37	INSULATION—EXTERNAL						
38	CASE						
39	NOZZLES						
40							
41	IGNITER, SAFE & ARM						
42	CASE						
43	CHARGE						
44	SAFE & ARM						
45	ATTACH. PROV.						
46							
47							
48							
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54							
55	COLUMN TOTALS						
56							
57	TOTAL—ENGINE INSTALLATION—SOLID						

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PAGE_____

NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

Propulsion Group-Engine Instal.—(con.)

DATE_____ **Air Breathing-Nuclear** REPORT_____

1	STEP						
2	SECTION						
3							
4	AIR BREATHING						
5	ENGINE						
6							
7	AFTERNURNER						
8							
9	GEARING & DRIVE						
10							
11	CONTROLS						
12							
13	AIR INDUCTION						
14							
15	EXHAUST SYSTEM						
16							
17	STARTING SYSTEM						
18							
19							
20							
21							
22	*SUPT. & INSTAL. STR.						
23							
24							
25							
26							
27							
28							
29	NUCLEAR *						
30	REACTOR						
31							
32	SHIELDING—THERMAL						
33							
34	NOZZLE						
35							
36	CONTROLS						
37							
38	HEAT EXCHANGER						
39							
40							
41							
42							
43							
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45							
46							
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48							
49							
50	SUPPORTS						
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—ENGINE INSTALLATION— AIRBREATHING—NUCLEAR						

* INCLUDE NACELLE, PYLON AND ENGINE SECTION NOT INTEGRAL WITH BODY

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

Propulsion Group-Propellant

DATE_____ **Fuel & Lub. Systems** REPORT_____

1	STEP						
2	SECTION						
3	TANKS, NON-STRUCTURAL						
4							
5	INTEG. TANK SEAL & SEALANT						
6	TANK SUPPORTS AND PADDING						
7	TANK BAFFLES						
8	TANK INSULATION						
9	TANK RELEASE & CONTROLS						
10	PUMP INSTALLATION						
11							
12							
13							
14							
15							
16	FILLING SYSTEM						
17	PLUMBING						
18	CONTROLS						
19	SUPPORTS						
20							
21	DISTRIBUTION SYSTEM						
22	PLUMBING						
23	CONTROLS						
24	SUPPORTS						
25							
26	TRANSFER SYSTEM						
27	PLUMBING						
28	CONTROLS						
29	SUPPORTS						
30							
31	VENT SYSTEM						
32	PLUMBING						
33	CONTROLS						
34	SUPPORTS						
35							
36	DRAINAGE SYSTEM						
37	PLUMBING						
38	CONTROLS						
39	SUPPORTS						
40							
41	PRESSURIZATION SYSTEM						
42	PLUMBING						
43	CONTROLS						
44	PRESSURE SOURCE						
45	HEAT EXCHANGER						
46	SUPPORTS						
47							
48	LUBRICATION SYSTEM						
49	TANKS						
50	PLUMBING						
51	COOLERS						
52	CONTROLS						
53	SUPPORTS						
54							
55	COLUMN TOTALS						
56							
57	TOTAL—FUEL & LUBRICATING SYSTEMS						

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PAGE_____

NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

Propulsion Group-Propellant

DATE_____ **Oxidizer & Lub. Systems** REPORT_____

1	STEP						
2	SECTION						
3	TANKS, NON-STRUCTURAL						
4							
5	INTEG. TANK SEAL & SEALANT						
6	TANK SUPPORTS AND PADDING						
7	TANK BAFFLES						
8	TANK INSULATION						
9	TANK RELEASE & CONTROLS						
10	PUMP INSTALLATION						
11							
12							
13							
14							
15							
16	FILLING SYSTEM						
17	PLUMBING						
18	CONTROLS						
19	SUPPORTS						
20							
21	DISTRIBUTION SYSTEM						
22	PLUMBING						
23	CONTROLS						
24	SUPPORTS						
25							
26	TRANSFER SYSTEM						
27	PLUMBING						
28	CONTROLS						
29	SUPPORTS						
30							
31	VENT SYSTEM						
32	PLUMBING						
33	CONTROLS						
34	SUPPORTS						
35							
36	DRAINAGE SYSTEM						
37	PLUMBING						
38	CONTROLS						
39	SUPPORTS						
40							
41	PRESSURIZATION SYSTEM						
42	PLUMBING						
43	CONTROLS						
44	PRESSURE SOURCE						
45	HEAT EXCHANGER						
46	SUPPORTS						
47							
48	LUBRICATION SYSTEM						
49	TANKS						
50	PLUMBING						
51	COOLERS						
52	CONTROLS						
53	SUPPORTS						
54							
55	COLUMN TOTALS						
56							
57	TOTALS—OXIDIZER & LUB. SYSTEMS						

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PAGE_____

NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____
Power Generating Group

DATE_____ **Prime Power Source** REPORT_____

1	STEP						
2	SECTION						
3							
4	CHEMICAL POWER UNIT*						
5	ENGINE						
6	ENGINE PUMP UNIT						
7	ENGINE GENERATOR UNIT						
8	GAS GENERATOR UNIT						
9	GAS GENERATOR						
10	FUEL STORAGE						
11	PLUMBING						
12	VALVES AND PUMPS						
13							
14							
15							
16	ELECTRIC POWER UNIT*						
17	BATTERY (QUAN.)						
18	WIRING						
19	BATTERY BOX & SUPPORTS						
20							
21							
22							
23							
24							
25							
26	NUCLEAR POWER UNIT*						
27	REACTOR						
28	SHIELDING						
29	HEAT EXCHANGER						
30	ENGINE						
31	ENGINE PUMP UNIT						
32	ENGINE GENERATOR UNIT						
33							
34							
35							
36							
37	COMPRESSED GAS SYSTEM*						
38	BOTTLE						
39	GAS						
40	PLUMBING AND VALVES						
41							
42							
43							
44	COOLING SYSTEM						
45	INTAKE & EXHAUST SYSTEM						
46	LUBRICATING SYSTEM						
47	STARTING SYSTEM						
48	SUPPORTS						
49	CONTROLS						
50							
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—PRIME POWER SOURCE						

*INDICATE WHERE GENERATOR OR PUMP IS INTEGRAL WITH THE POWER SOURCE

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

Power Generating Group—(con.)

DATE_____ **Hydraulic and Pneumatic** REPORT_____

1	STEP						
2	SECTION						
3							
4	HYDRAULIC SYSTEMS						
5	MOTOR PUMP UNIT						
6	PUMP UNIT						
7	RESERVOIRS						
8	ACCUMULATORS						
9	FILTERS						
10	PRESSURE REGULATORS						
11	VALVES						
12	CONTROLS						
13	PLUMBING						
14	FLUID						
15							
16							
17							
18							
19							
20							
21	INSULATION						
22							
23							
24	EXTERNAL SERVICE PROV.						
25							
26	SUPPORTS						
27							
28	PNEUMATIC SYSTEMS						
29	MOTOR COMPRESSOR UNIT						
30	COMPRESSOR UNIT						
31	RESERVOIRS						
32	ACCUMULATORS						
33	FILT. & WATER SEPARATORS						
34	PRESSURE REGULATORS						
35	VALVES						
36	CONTROLS						
37	PLUMBING						
38	FLUID						
39							
40							
41							
42							
43							
44							
45							
46							
47							
48	INSULATION						
49							
50							
51	EXTERNAL SERVICE PROV.						
52							
53	SUPPORTS						
54							
55	COLUMN TOTALS						
56							
57	TOTAL—HYDRAULIC & PNEUMATIC						

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NAME_____ PAGE_____

DATE_____ MODEL_____

REPORT_____

DETAIL WEIGHT STATEMENT
Power Generating Group—(con.)
Electrical

1	STEP						
2	SECTION						
3							
4	A.C. ELECTRICAL SYSTEMS						
5							
6	GENERATORS						
7	TRANSFORMERS						
8							
9	INVERTERS DC TO AC						
10	PHASE ADAPTERS						
11	FREQUENCY CONVERTERS						
12	VOLTAGE REGULATORS						
13	CONSTANT SPEED DRIVES						
14							
15							
16	LIGHTS						
17	SIGNAL DEVICES						
18	CONTROLS						
19							
20	WIRING & PLUGS						
21							
22	RELAYS						
23							
24	CIRCUITRY EXCL. WIRING						
25							
26	EXTERNAL SERVICE PROV.						
27							
28	SUPPORTS						
29							
30	D.C. ELECTRICAL SYSTEMS						
31							
32	GENERATORS						
33	CONVERTERS AC TO DC						
34	VOLTAGE REGULATORS						
35	RECTIFIERS						
36							
37							
38	LIGHTS						
39	SIGNAL DEVICES						
40	CONTROLS						
41							
42	WIRING & PLUGS						
43							
44	RELAYS						
45							
46							
47							
48	CIRCUITRY EXCL. WIRING						
49							
50	EXTERNAL SERVICE PROV.						
51							
52	SUPPORTS						
53							
54							
55	COLUMN TOTALS						
56	TOTAL ELECTRICAL						
57	TOTAL—POWER GENERATOR GROUP						

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

Orientation Controls Group

DATE_____ **Aerodynamic** REPORT_____

1	STEP						
2	SECTION						
3							
4	AERODYNAMIC CONTROL—ROLL						
5	TYPE*						
6	ACTUATOR & VALVE						
7	MISC. VALVES, RELAYS, ETC.						
8	PLUMBING OR WIRE						
9	FLUID						
10	ELECTRICAL/MECHANICAL						
11	INDIVIDUAL POWER UNIT						
12							
13							
14							
15							
16	SUPPORTS						
17							
18	AERODYNAMIC CONTROL—PITCH						
19	TYPE*						
20	ACTUATOR & VALVE						
21	MISC. VALVES, RELAYS, ETC.						
22	PLUMBING OR WIRE						
23	FLUID						
24	ELECTRICAL/MECHANICAL						
25	INDIVIDUAL POWER UNIT						
26							
27							
28							
29							
30	SUPPORTS						
31							
32	AERODYNAMIC CONTROL—YAW						
33	TYPE*						
34	ACTUATOR & VALVE						
35	MISC. VALVES, RELAYS, ETC.						
36	PLUMBING OR WIRING						
37	FLUID						
38	ELECTRICAL/MECHANICAL						
39	INDIVIDUAL POWER UNIT						
40							
41							
42							
43	SUPPORTS						
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54							
55	COLUMN TOTALS						
56							
57	TOTAL						

* HYD—HYDRAULIC, PNEU—PNEUMATIC, ELECT—ELECTRIC

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

Orientation Controls Group—(con.)

DATE_____ **Thrust and Reaction** REPORT_____

1 STEP						
2 SECTION						
3 MAIN THRUST DIRECTION						
4 TYPE*						
5 ACTUATORS & VALVES						
6 MISC. VALVES, RELAYS, ETC.						
7 PLUMBING OR WIRING						
8 FLUID						
9 ELECTRICAL/MECHANICAL						
10 INDIVIDUAL POWER UNIT						
11						
12						
13						
14						
15 SUPPORTS						
16						
17 AUX. THRUST DIRECTION						
18 TYPE*						
19 ACTUATORS & VALVES						
20 MISC. VALVES, RELAYS, ETC.						
21 PLUMBING OR WIRING						
22 FLUID						
23 ELECTRICAL/MECHANICAL						
24 INDIVIDUAL POWER UNIT						
25						
26						
27						
28 SUPPORTS						
29						
30 REACTION CONTROL SYSTEM						
31 ROLL						
32						
33						
34						
35						
36 PITCH						
37						
38						
39						
40						
41						
42						
43						
44						
45						
46						
47						
48 YAW						
49 VELOCITY CONTROL SYSTEMS						
50 VERNIER ENGINES						
51						
52						
53						
54						
55 COLUMN TOTALS						
56						
57 TOTAL—ORIENTATION CONTROLS GROUP						

* HYD.--HYDRAULIC, PNEU PNEUMATIC, ELECT ELECTRIC

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Guidance Group** REPORT_____

1	STEP						
2	SECTION						
3							
4	GUIDANCE SOURCE						
5	LAUNCH TYPE*						
6							
7							
8							
9							
10							
11	MID COURSE TYPE*						
12							
13							
14							
15							
16							
17							
18	TERMINAL TYPE*						
19							
20							
21							
22							
23							
24							
25							
26	PRIME REFERENCE						
27	GYRO STABLE PLATFORM						
28	GYRO ELECTRONICS						
29	COMPUTER						
30							
31							
32							
33	AUX. INTELLIGENCE						
34	ALTITUDE						
35	VELOCITY						
36							
37							
38	POWER SUPPLY						
39	CIRCUITRY						
40	SUPPORTS						
41							
42							
43	OUTPUT—A.F.C. SYSTEM						
44	ELECTRONIC PACKAGE						
45	RATE GYROS						
46	ACCELEROMETERS						
47	SERVO AMPLIFIERS						
48	POWER SUPPLY						
49	CIRCUITRY						
50	SUPPORTS						
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—GUIDANCE GROUP						

* P—PROGRAMER, C—RADIO COMMAND, R—RADAR, I—INFRARED, M—MAPMACH

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Electronics Group** REPORT_____

1	STEP						
2	SECTION						
3							
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49							
50	WIRING & PLUGS—IF NOT LISTED						
51	ABOVE						
52	RACKS & SUPPORTS						
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—ELECTRONICS GROUP						

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Environmental Protection Group** REPORT_____

1 STEP						
2 SECTION						
3						
4						
5 TEMP. CONTROL—STRUCTURE						
6 HEAT EXCHANGERS						
7 PUMPS						
8 BLOWERS						
9 DUCTS						
10 TANKAGE						
11 PLUMBING						
12 FLUID						
13						
14						
15 SUPPORTS						
16						
17						
18						
19 TEMP. CONTROL—EQUIPMENT						
20 HEAT EXCHANGERS						
21 PUMPS						
22 BLOWERS						
23 DUCTS						
24 TANKAGE						
25 PLUMBING						
26 FLUID						
27						
28						
29 SUPPORTS						
30						
31						
32						
33 COMPARTMENT PRESS. SYSTEM						
34						
35						
36						
37						
38						
39						
40						
41						
42 GROUND SERVICE PROVISIONS						
43 HEATING						
44						
45 COOLING						
46						
47 ATMOSPHERE CONTROL						
48						
49						
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51						
52						
53						
54						
55 COLUMN TOTALS						
56						
57 TOTAL—ENVIRONMENTAL PROTECTION GROUP						

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Armament Group** REPORT_____

1	STEP						
2	SECTION						
3							
4	SAFETY AND ARMING						
5	DEVICES						
6							
7							
8	SUPPORTS						
9							
10							
11							
12	DETONATING						
13	DEVICES						
14							
15							
16	SUPPORTS						
17							
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31							
32	WARHEAD PROVISION						
33	SUPPORTS						
34	CHECKS						
35	ACCESSORIES						
36							
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55	COLUMN TOTAL						
56							
57	TOTAL ARMAMENT						

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NAME_____ **DETAIL WEIGHT STATEMENT** MODEL_____

DATE_____ **Miscellaneous Group** REPORT_____

1	STEP						
2	SECTION						
3							
4	STEP SEPARATION SYSTEMS						
5	SEPARATION GUIDES						
6							
7	SEPARATION ROCKETS						
8	MOTORS—EMPTY						
9	INSTALLATION						
10							
11	EXPLOSIVE DEVICES						
12							
13	GAS INITIATION SYSTEMS						
14	CONT. & SEQUENCING EQUIP.						
15							
16	CIRCUITRY						
17							
18							
19	SUPPORTS, COMMON						
20							
21							
22	DESTRUCT SYSTEMS						
23	DESTRUCT COMMAND REC.						
24	RANGE SAFETY BEACONS						
25							
26	DESTRUCT POWER SUPPLY						
27	DESTRUCT CONTROLS						
28							
29	AUTO. DESTRUCT DEVICES						
30							
31	EXPLOSIVES						
32							
33							
34	SUPPORTS—COMMON						
35							
36							
37	EMERGENCY EQUIPMENT						
38	MALFUNCTION DETECTION						
39							
40							
41							
42							
43	FIRE EXTINGUISHING						
44							
45							
46							
47							
48	VISUAL IDENTIFICATION						
49							
50							
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL—MISCELLANEOUS GROUP						

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APPENDIX I

Supplementary Data and *Structural Increments for Design Features

NAME_____ DATE_____ PAGE_____ MODEL_____ REPORT_____

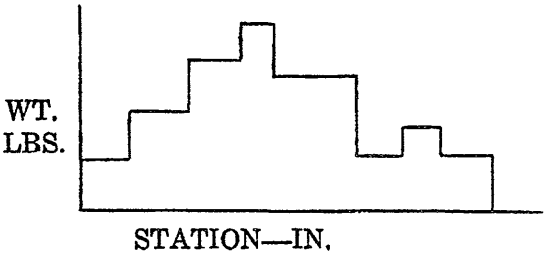
1	STEP						
2	SECTION						
3	BODY—JOINTS						
4	(STA)						
5	(STA)						
6	(STA)						
7							
8	CUTOUTS—BLAST PORTS						
9	—						
10	—						
11	—						
12	—						
13	REMOVABLE PANELS—STRUCT.						
14	—ELECTRONIC						
15							
16	—ENGINE						
17	—FUEL TANK						
18	—						
19	—						
20	FUEL TANK—INTEGRAL						
21	—						
22	—						
23	—PRESSURIZATION						
24							
25							
26							
27	TEMPERATURE REQUIREMENT						
28	—WING						
29	—SURFACES						
30	—BODY						
31	—						
32	—						
33							
34							
35	DUPLICATE SYS. (RELIABILITY)						
36	—PROPULSION						
37	—ELECTRICAL						
38	—HYDRAULIC						
39	—ORIENTATION CONT.						
40	—TEMP. CONTROLS						
41							
42							
43	STATIC LOADS						
44							
45							
46							
47							
48							
49							
50							
51							
52							
53							
54							
55	COLUMN TOTALS						
56							
57	TOTAL						

* ALREADY INCLUDED IN STRUCTURAL GROUPS. USE ADDITIONAL PAGE IF NECESSARY. SEE PAGE II FOR INSTRUCTIONS AND NOTES.

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INSTRUCTIONS AND NOTES—ASSUME NO CHANGE IN BASIC CONFIGURATION, TYPE OF CONSTRUCTION OR DESIGN CONDITIONS. OVERLAPPING PENALTIES SHALL BE EXPLAINED BY APPROPRIATE DISCUSSION BELOW. ANY KNOWN WEIGHT Increments OR DECREMENTS FOR UNUSUAL DESIGN REQUIREMENTS, SUCH AS RIGIDITY, FATIGUE, TRANSIENT LOADS, OR USE OF HIGH STRENGTH OR SPECIAL MATERIALS. SHALL BE DISCUSSED.

DISCUSSION



Indicate by graphic plot (as above) the weight distribution of the structural elements of the missile along the longitudinal axis (Stations.). Also identify by appropriate nomenclature the points of large change or irregularities in the weight distribution.

ADD SUPPLEMENTAL PAGE IF NECESSARY

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PART III

WEIGHT AND BALANCE STATUS

GUIDED MISSILE

AND

SPACE LAUNCH VEHICLES

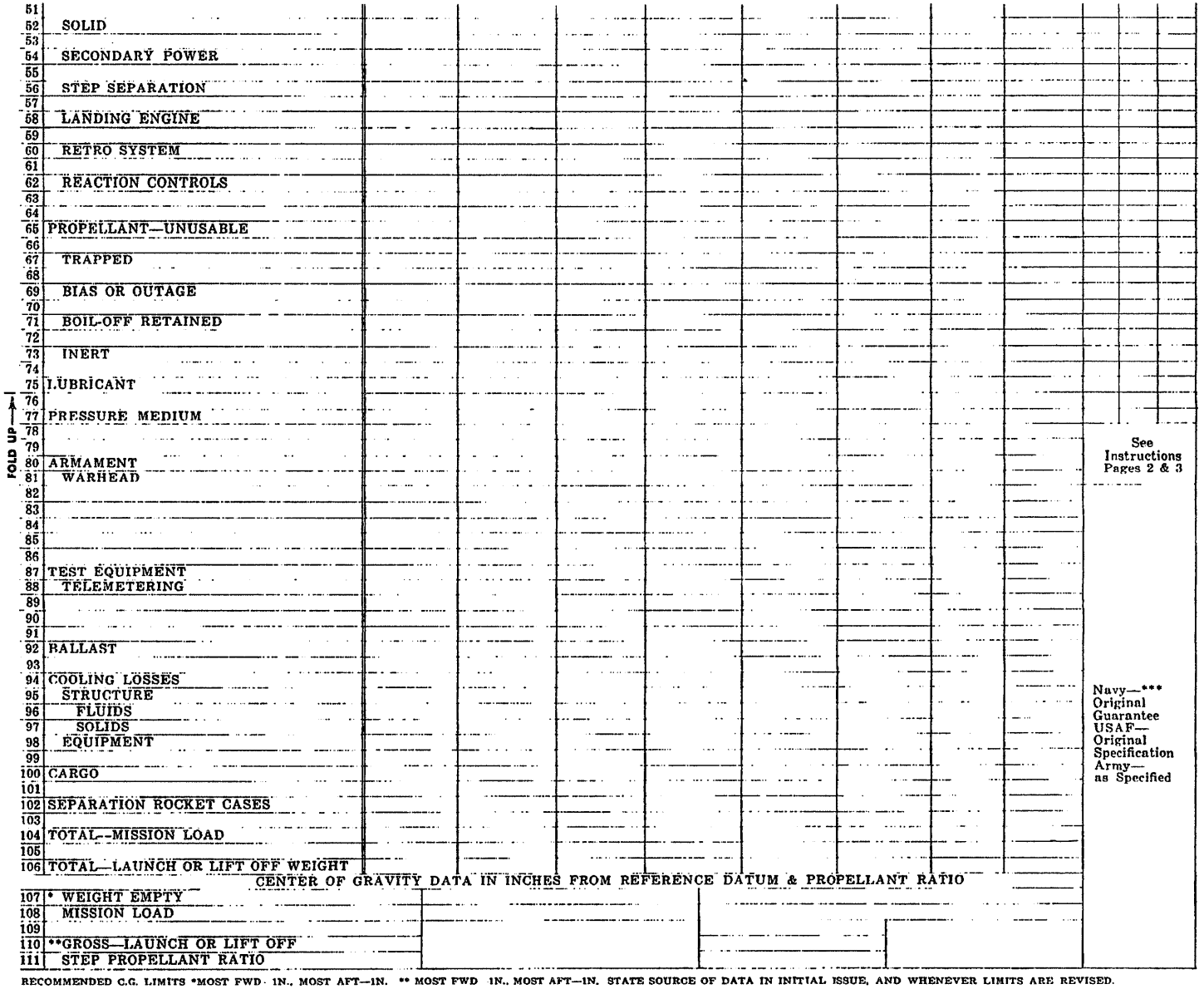
Note: Refer to par 4.9. Full size copies printed on vellum paper suitable for ozalid, blueprint or similar methods of reproduction may be obtained from the procuring activity or as directed by the contracting officer. Such requests should specify "Full size vellum form PART III of MIL-STD-176."

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	Model Engine Contract Number Date Issue	* * * Original Weight As of Dated	Gov't Responsibility Changes Auth, Pending & GFE Variations	Revised Weight Sum Columns 2 & 3	Total Contractor Responsibility Changes & Variations Column 6 minus Column 4	Current Status Sum Columns 4 & 5 and Sum Columns 7, 8 & 9	Last Status Dated Issue	Change between last & current Status Reports		Basis for Current Data		
								Ttotal Gov't Responsibility Changes & GFE Variations	Total Contractor Responsibility Changes	% EST	% CALC	% ACT
	1	2	3	4	5	6	7	8	9	10	11	12
1	AERODYNAMIC SURFACES											
2												
3												
4												
5	BODY											
6												
7												
8												
9	TAKE-OFF AND RECOVERY											
10												
11	PROPULSION											
12	ENGINE INSTALLATION											
13	PROPELLANT SYSTEMS											
14												
15												
16	POWER GENERATING SYSTEMS											
17	PRIME POWER SOURCE											
18	HYDRAULIC & PNEUMATIC											
19	ELECTRICAL											
20												
21	ORIENTATION CONTROLS											
22												
23	GUIDANCE											
24												
25	ELECTRONICS											
26												
27	ENVIRONMENTAL PROTECTION											
28												
29	ARMAMENT											
30												
31	MISCELLANEOUS											
32	STEP SEPARATION SYSTEMS											
33	DESTRUCT SYSTEMS											
34	EMERGENCY EQUIPMENT											
35	VISUAL IDENTIFICATION											
36												
37												
38												
39												
40												
41												
42	TOTAL—CONTRACTOR CONTROLLED											
43	TOTAL—GFE											
44	TOTAL—WEIGHT EMPTY											
45												
46												
47	PROPELLANT—USABLE											
48	LIQUID FUEL											
49	OXIDIZER											
50												

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	Model Date Issue	Government Responsibility Changes and GFE Variations Since Last Report											Totals Columns 2 Through 12
		Changes										GFE VAR.	
	1	2	3	4	5	6	7	8	9	10	11	12	13
1	AERODYNAMIC SURFACES												
2													
3													
4													
5	BODY												
6													
7													
8													
9	TAKE-OFF AND RECOVERY												
10													
11	PROPULSION												
12	ENGINE INSTALLATION												
13	PROPELLANT SYSTEMS												
14													
15													
16	POWER GENERATING SYSTEMS												
17	PRIME POWER SOURCE												
18	HYDRAULIC & PNEUMATIC												
19	ELECTRICAL												
20													
21	ORIENTATION CONTROLS												
22													
23	GUIDANCE												
24													
25	ELECTRONICS												
26													
27	ENVIRONMENTAL PROTECTION												
28													
29	ARMAMENT												
30													
31	MISCELLANEOUS												
32	STEP SEPARATION SYSTEMS												
33	DESTRUCT SYSTEMS												
34	EMERGENCY EQUIPMENT												
35	VISUAL IDENTIFICATION												
36													
37													
38													
39													
40													
41													
42	TOTAL—CONTRACTOR CONTROLLED												
43	TOTAL—GFE												
44	TOTAL—WEIGHT EMPTY												
45													
46													
47	PROPELLANT—USABLE												
48	LIQUID FUEL												
49	OXIDIZER												
50													
51													
52	SOLID												
53													

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54	SECONDARY POWER													
55														
56	STEP SEPARATION													
57														
58	LANDING ENGINE													
59														
60	RETRO SYSTEM													
61														
62	REACTION CONTROLS													
63														
64														
65	PROPELLANT—UNUSABLE													
66														
67	TRAPPED													
68														
69	BIAS OR OUTAGE													
70														
71	BOIL-OFF RETAINED													
72														
73	INERT													
74														
75	LUBRICANT													
76														
77	PRESSURE MEDIUM													
78														
79														
80	ARMAMENT													
81	WARHEAD													
82														
83														
84														
85														
86														
87	TEST EQUIPMENT													
88	TELEMETERING													
89														
90														
91														
92	BALLAST													
93														
94	COOLING LOSSES													
95	STRUCTURE													
96	FLUIDS													
97	SOLIDS													
98	EQUIPMENT													
99														
100	CARGO													
101														
102	SEPARATION ROCKET CASES													
103														
104	TOTAL—MISSION LOAD													
105														
106	TOTAL—LAUNCH OR LIFT OFF WEIGHT													

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INSTRUCTIONS

1. ALL ENTRIES ON PAGE 1 COL. 8 SHALL BE DETAILED IN THE TABLE PAGE 2 USING A SEPARATE COLUMN FOR EACH CHANGE AND ADDITIONAL SHEETS IF NECESSARY. CHANGES ARE TO BE IDENTIFIED IN TABULAR SUMMARY CALLED FOR IN INSTRUCTION 2.

2. ATTACH TABULAR SUMMARY OF GOV'T. RESPONSIBILITY CHANGES INCLUDED ON PAGE 1 COL. 3 LISTING, AS APPLICABLE, THE FOLLOWING INFORMATION—CONTRACT CHANGE ORDER NUMBER OR LETTER DESIGNATION, CORRESPONDENCE REFERENCE FOR PENDING CHANGES, BRIEF SUBJECT TITLE, AND CHANGES IN WEIGHT, HORIZONTAL MOMENT, OR C.G. IN INCHES FOR EACH OF THE FOLLOWING CONDITIONS-WEIGHT EMPTY, MISSION LOAD, SEPARABLE BOOSTER AND LAUNCHING GROSS WEIGHT. FOR GOV'T. RESPONSIBILITY CHANGES ADDED SINCE LAST REPORT AND SUMMARIZED ON PAGE 1 COL. 8, LIST THE COLUMN NUMBER OF PAGE 2 IN WHICH THE WEIGHT CHANGE IS DETAILED. AUTHORIZED CHANGES INCLUDED ON PAGE 1 COL. 3 SHALL BE LISTED AS A SEPARATE GROUP, AND MAY REFERENCE AND CARRY FORWARD THE TOTAL OF WEIGHTS, MOMENTS, AND C.G. CHANGES FOR THE AUTHORIZED CHANGES SUMMARIZED IN THE LAST STATUS REPORT.

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	Model Date Issue	Contractor Responsibilty Changes Since Last Report											Totals Columns 2 Through 12
		Changes											
	Contractor Change Identification												
	1	2	3	4	5	6	7	8	9	10	11	12	13
1	AERODYNAMIC SURFACES												
2													
3													
4													
5	BODY												
6													
7													
8													
9	TAKE-OFF AND RECOVERY												
10													
11	PROPULSION												
12	ENGINE INSTALLATION												
13	PROPELLANT SYSTEMS												
14													
15													
16	POWER GENERATING SYSTEMS												
17	PRIME POWER SOURCE												
18	HYDRAULIC & PNEUMATIC												
19	ELECTRICAL												
20													
21	ORIENTATION CONTROLS												
22													
23	GUIDANCE												
24													
25	ELECTRONICS												
26													
27	ENVIRONMENTAL PROTECTION												
28													
29	ARMAMENT												
30													
31	MISCELLANEOUS												
32	STEP SEPARATION SYSTEMS												
33	DESTRUCT SYSTEMS												
34	EMERGENCY EQUIPMENT												
35	VISUAL IDENTIFICATION												
36													
37													
38													
39													
40													
41													
42	TOTAL—CONTRACTOR CONTROLLED												
43	TOTAL—GFE												
44	TOTAL—WEIGHT EMPTY												
45													
46													
47	PROPELLANT—USABLE												
48	LIQUID FUEL												
49	OXIDIZER												
50													
51													
52	SOLID												
53													

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54	SECONDARY POWER													
55														
56	STEP SEPARATION													
57														
58	LANDING ENGINE													
59														
60	RETRO SYSTEM													
61														
62	REACTION CONTROLS													
63														
64														
65	PROPELLANT—UNUSABLE													
66														
67	TRAPPED													
68														
69	BIAS OR OUTAGE													
70														
71	BOIL-OFF RETAINED													
72														
73	INERT													
74														
75	LUBRICANT													
76														
77	PRESSURE MEDIUM													
78														
79														
80	ARMAMENT													
81	WARHEAD													
82														
83														
84														
85														
86														
87	TEST EQUIPMENT													
88	TELEMETERING													
89														
90														
91														
92	BALLAST													
93														
94	COOLING LOSSES													
95	STRUCTURE													
96	FLUIDS													
97	SOLIDS													
98	EQUIPMENT													
99														
100	CARGO													
101														
102	SEPARATION ROCKET CASES													
103														
104	TOTAL—MISSION LOAD													
105														
106	TOTAL—LAUNCH OR LIFT OFF WEIGHT													

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INSTRUCTION CONTINUED FROM PAGE 2

3. ALL ENTRIES ON PAGE 1 COL. 0 SHALL BE DETAILED IN THE TABLE ON PAGE 3 USING A SEPARATE COLUMN FOR EACH CHANGE AND ADDITIONAL SHEETS IF NECESSARY. MINOR CHANGES IN GROUP WEIGHTS DUE TO REVISED ESTIMATED, CALCULATED, OR ACTUAL WEIGHTS NEED NOT BE LISTED IN DETAIL AND MAY BE COMBINED INTO ONE OR MORE COLUMNS USING APPROPRIATE HEADING SUCH AS—CALC VS EST, ETC. INCLUDE BRIEF DISCUSSION OF REASONS FOR ALL IMPORTANT OR MAJOR WEIGHT CHANGES, VARIATIONS, OR C.G. SHIFTS, INCLUDING HORIZONTAL MOMENT CHANGES AND ANY REMEDIAL MEASURES BEING INVESTIGATED OR APPLIED, USING COLUMN NUMBERS FROM PAGE 3 AND BRIEF CHANGE TITLES TO IDENTIFY EACH CHANGE.

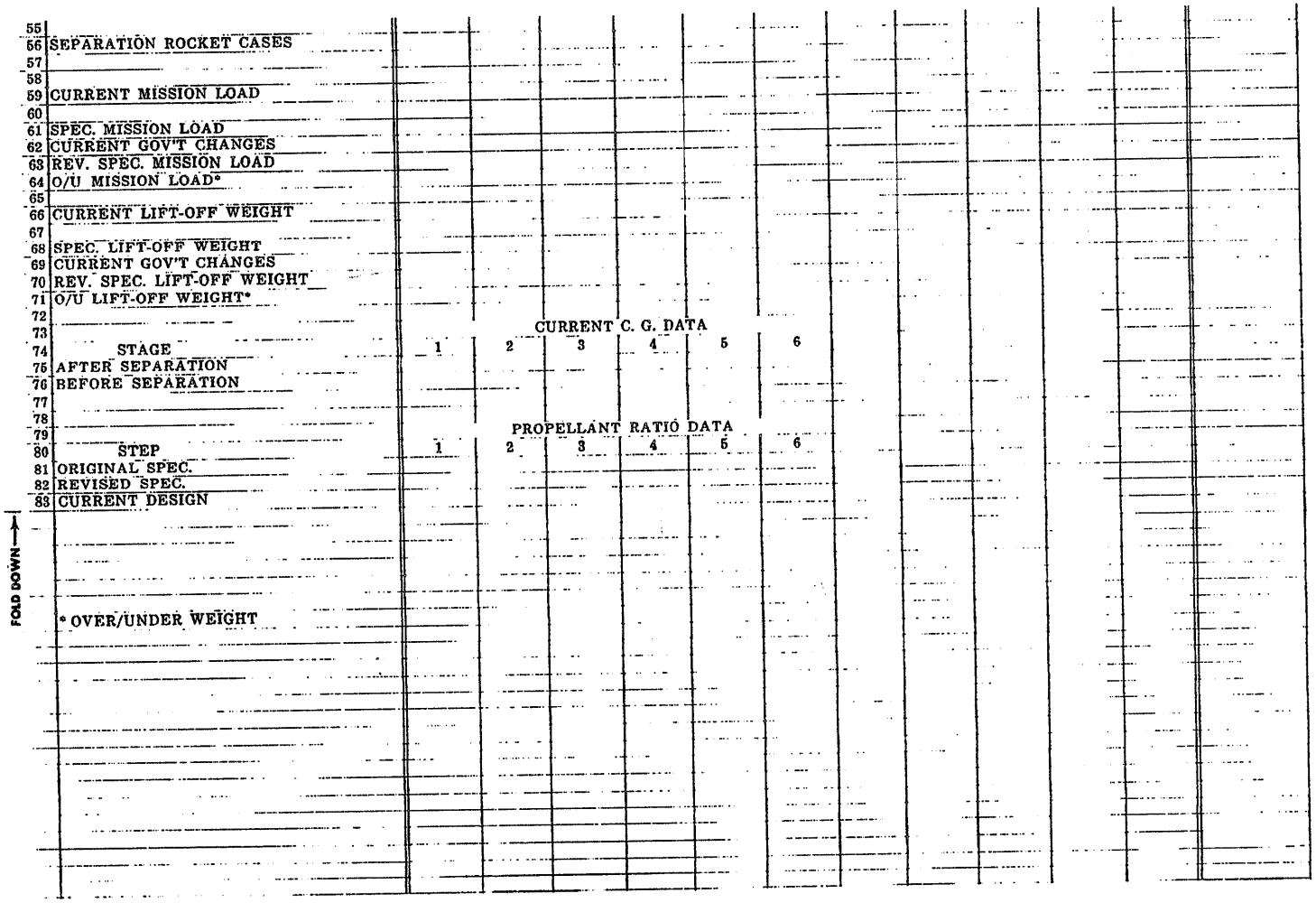
4. ON PAGE 1 FOR EACH ITEM OF WEIGHT EMPTY, UNDER BASIS FOR CURRENT WEIGHT DATA, COLS. 10-12, LIST PERCENTAGES OF CURRENT GROUP WEIGHTS OF COL. 6 WHICH ARE IN AN ESTIMATED, CALCULATED, OR ACTUAL WEIGHT STATUS.

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	Model Date Issue	Weight and Balance Status Guided Missile & Space Launch Vehicle											
												All Steps	
1	STEP												
2	SECTION												
3	—WEIGHT EMPTY—												
4	AERODYNAMIC SURFACES												
5	BODY												
6	TAKE OFF AND RECOVERY												
7	ENGINE INSTALLATION												
8	PROPELLANT SYSTEMS												
9	PRIME POWER SOURCE												
10	HYDRAULIC & PNEUMATIC												
11	ELECTRICAL												
12	ORIENTATION CONTROLS												
13	GUIDANCE												
14	ELECTRONICS												
15	ENVIRONMENTAL PROTECTION												
16	ARMAMENT												
17	STEP SEPARATION SYSTEMS												
18	DESTRUCT SYSTEMS												
19	EMERGENCY EQUIPMENT												
20	VISUAL IDENTIFICATION												
21													
22													
23													
24	CURRENT WEIGHT EMPTY												
25													
26	SPEC. WEIGHT EMPTY												
27	CURRENT GOV'T CHANGES												
28	REV. SPEC. WEIGHT EMPTY												
29	O/U WEIGHT EMPTY*												
30													
31													
32	MISSION LOAD												
33													
34	LIQUID FUEL—USABLE												
35	OXIDIZER—USABLE												
36													
37	SOLID PROPELLANT												
38	SECONDARY POWER PROPELLANT												
39	STEP SEPARATION PROPELLANT												
40	LANDING ENGINE PROPELLANT												
41	RETRO SYSTEM PROPELLANT												
42	REACTION CONTROL PROPELLANT												
43	TRAPPED PROPELLANT												
44	BIAS OR OUTAGE PROPELLANT												
45	BOIL-OFF RETAINED PROPELLANT												
46	INERT PROPELLANT												
47	LUBRICANT												
48	PRESSURE MEDIUM												
49	ARMAMENT												
50	TEST EQUIPMENT												
51	BALLAST												
52	COOLING LOSSES												
53													
54	CARGO												

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